BioHackrXiv

BioHackEU26 report for Project 10: Enabling workflows and EDAM annotations to display in community website deployments

[1], [1], **Ove Johan Ragnar Gustafsson** [1], **and** [1]

1, **1**, **1** Australian BioCommons, University of Melbourne, Melbourne, Victoria, Australia **1**

BioHackathon series:
BioHackathon Europe 2026
Barcelona, Spain, 2026
Project 10

Submitted: 11 Nov 2026

License:
Authors retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC-BY).

Published by BioHackrXiv.org

Note: Please keep sections to a maximum of only two levels.

Introduction

Project abstract from BioHackathon:

The ELIXIR Toolkit Theme (ETT, https://elixir-belgium.github.io/elixir-toolkit-theme/) enables communities to deploy websites via GitHub, facilitating broad contributions and collaborations, and streamlining the process of building and populating the site. The popularity of the ETT is demonstrated by numerous deployments, including RDMkit, RSQkit, and WorkflowHub (Gustafsson et al., 2025) documentation. These deployments can also centralise sharing of digital resources such as workflows intended for reuse by communities. This use case requires a streamlined way to find, view, and understand workflows, in the context of an ETT deployed website.

Whilst ETT already offers the possibility of annotating pages with identifiers from FAIRsharing and bio.tools (Mendes et al., 2026), support for workflows and ontologies is not implemented. This project will extend the ETT to enable 1) automation of workflow imports from WorkflowHub, 2) creation of interactive tables to present the communities workflows, and 3) automated addition of EDAM ontology annotations at both the workflow table and webpage (i.e. schema.org) level. The latter will build on the existing integration with bio.tools, offer the possibility of manual curation, and increase the findability of these webpages. Moreover, ontology terms (e.g. EDAM) are already used to annotate the bio.tools elements already included on many ETT instances, including RDMkit.

Workflow developers from any community can thus continue using WorkflowHub to make their workflows FAIR, and community coordinators can independently and automatically import these to an ETT deployment along with EDAM annotations. This support would be beneficial for increasing the findability of the pages and enabling better integration across multiple instances of ETT.

Results

Project goals

1. Translate a proof of concept integration with the WorkflowHub API to the ELIXIR Toolkit Theme, including GitHub actions, configs, htmls, and css.
2. Enable support for annotation with ontology terms on ETT, including import of ontology annotations from bio.tools based on existing annotations of pages.
3. Create working examples, including documentation, that can be included on the ETT test deployment page: e.g. https://elixir-belgium.github.io/elixir-toolkit-theme/all_trainings

WorkflowHub import

Use case driver https://australian-structural-biology-computing.github.io/index

Idea for table include https://australianbiocommons.github.io/workflowfinder/

Source collection on WorkflowHub: https://workflowhub.eu/collections/40

EDAM annotations

Discussion

Acknowledgements

The authors acknowledge all the organisers and participants at the ELIXIR BioHackathon Europe 2026 in Barcelona, Spain, for helping make this project a success.

This work is supported by Australian BioCommons, which is enabled by NCRIS via Bioplatforms Australia funding.

References

Author contributions

Ove Johan Ragnar Gustafsson: Conceptualization, writing – original draft.

References

Gustafsson, O. J. R., Wilkinson, S. R., Bacall, F., Soiland-Reyes, S., Leo, S., Pireddu, L., Owen, S., Juty, N., Fernández, J. M., Brown, T., et al. (2025). Workflowhub: A registry for computational workflows. *Scientific Data*, *12*(1), 837 **[cito:citesAsAuthority]**.

Mendes, A. I., Ienasescu, H., Andersen, C. A., Calhoun, E., Chmura, P. J., Collier, K., Goga, I., Gustafsson, O. J., Kalaš, M., Kierkegaard, M., et al. (2026). Bio. tools: An expanded web service for research software in the life sciences. *Nucleic Acids Research*, *54*(W1), W4–W12 **[cito:citesAsAuthority]**.

Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the source to see the exact structure.